Ireland's Electricity Grid 2022–2025: Is Wind Keeping Up with Demand?

YEAR

2022 | 2023 | 2024 | 2025

Total Electricity Demand (TWh)

132.45

Demand Met by Wind

33.8%

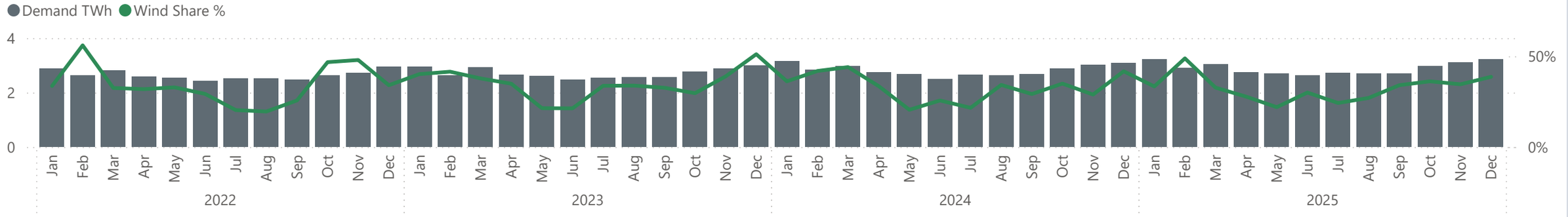
Available Wind Wasted

11.2%

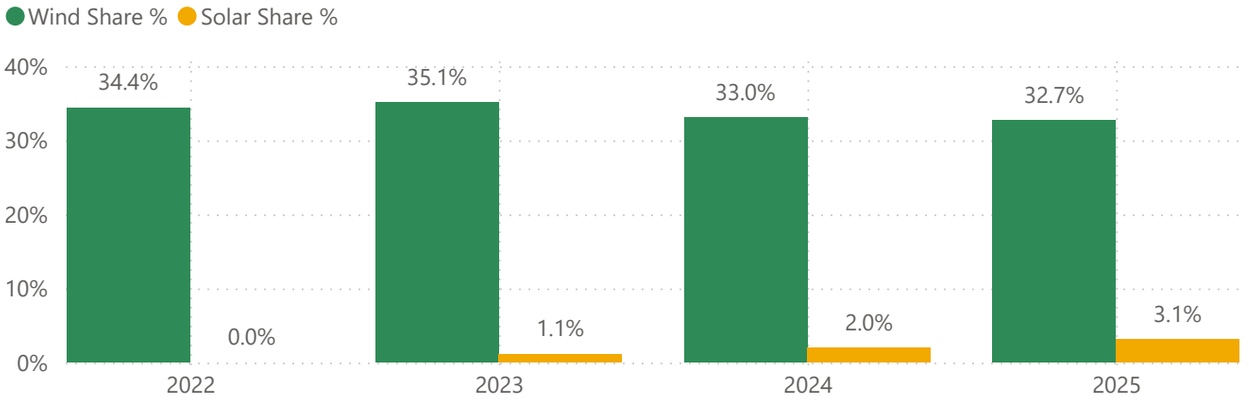
Highest Demand Recorded (MW)

6,012

Monthly Demand (TWh) vs Wind Share (%)



Wind and Solar Share of Demand by Year



2022–2025: Demand grew 9.5% from 2022 to 2025, but wind output stayed flat. Wind's share fell from 35% to 33%, and 13% of available wind was wasted in 2025.

When Does Ireland Need the Most Electricity?

YEAR

2022 | 2023 | 2024 | 2025

SEASON

Autumn | Spring | Summer | Winter

Peak Demand (MW)

6.012

Average Demand (MW)

3,777

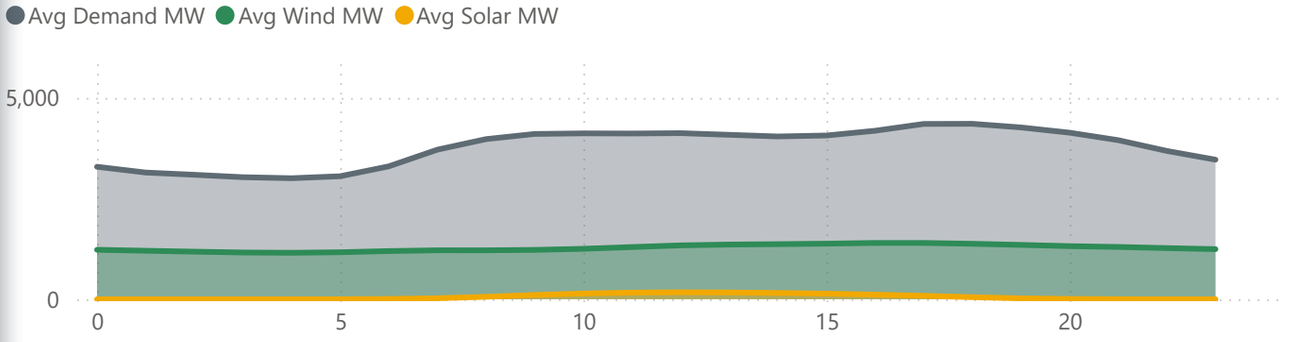
Demand Growth vs Last Year

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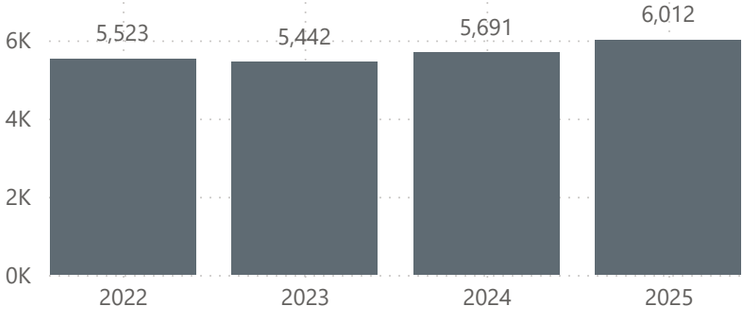
Demand Heatmap: Hour of Day × Month

hour_of_day	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0	3,538	3,507	3,474	3,280	3,096	3,078	3,068	3,048	3,078	3,225	3,483	3,610
1	3,366	3,340	3,323	3,157	2,969	2,941	2,935	2,925	2,962	3,097	3,328	3,432
2	3,280	3,267	3,253	3,089	2,916	2,888	2,887	2,883	2,929	3,056	3,283	3,370
3	3,213	3,203	3,190	3,025	2,861	2,835	2,835	2,832	2,882	2,997	3,217	3,290
4	3,179	3,176	3,165	3,009	2,847	2,811	2,817	2,814	2,866	2,971	3,180	3,239
5	3,235	3,242	3,230	3,077	2,880	2,816	2,831	2,862	2,935	3,037	3,241	3,283
6	3,497	3,525	3,499	3,305	3,112	3,039	3,027	3,036	3,202	3,314	3,534	3,527
7	3,935	3,977	3,922	3,705	3,547	3,420	3,378	3,378	3,632	3,771	4,005	3,926
8	4,175	4,202	4,161	3,997	3,812	3,706	3,692	3,686	3,883	4,025	4,236	4,159
9	4,364	4,363	4,282	4,083	3,883	3,807	3,835	3,834	3,969	4,126	4,375	4,350
10	4,449	4,391	4,278	4,064	3,857	3,810	3,849	3,849	3,950	4,121	4,388	4,436
11	4,485	4,395	4,264	4,040	3,829	3,805	3,854	3,844	3,932	4,106	4,383	4,467
12	4,518	4,413	4,268	4,032	3,823	3,807	3,865	3,851	3,933	4,106	4,396	4,497
13	4,489	4,371	4,220	3,975	3,772	3,759	3,824	3,806	3,885	4,062	4,365	4,482
14	4,453	4,326	4,171	3,911	3,726	3,717	3,773	3,755	3,842	4,023	4,360	4,464
15	4,498	4,357	4,192	3,906	3,740	3,726	3,761	3,753	3,865	4,051	4,439	4,521
16	4,676	4,476	4,302	3,984	3,824	3,795	3,821	3,822	3,962	4,163	4,657	4,740
17	4,969	4,702	4,467	4,118	3,953	3,910	3,935	3,936	4,094	4,321	4,930	4,924
18	4,919	4,838	4,570	4,131	3,955	3,900	3,919	3,921	4,093	4,386	4,848	4,823
19	4,724	4,697	4,597	4,079	3,885	3,816	3,819	3,834	4,073	4,383	4,660	4,653
20	4,504	4,473	4,430	4,070	3,808	3,724	3,714	3,764	4,060	4,195	4,448	4,465

A Typical Day: Average Demand, Wind and Solar by Hour



Annual Peak Demand (MW)



Annual peak demand always occurs on a winter weekday around 17:30 and grew 8.9% from 2022 to 2025. Wind output stays almost flat through the day, and solar fades before the evening peak.

Is Ireland Making the Most of Its Wind?

YEAR

2022 | 2023 | 2024 | 2025

SEASON

Autumn | Spring | Summer | Winter

Demand Met by Wind

33.8%

Wind Wasted (% of Available)

11.2%

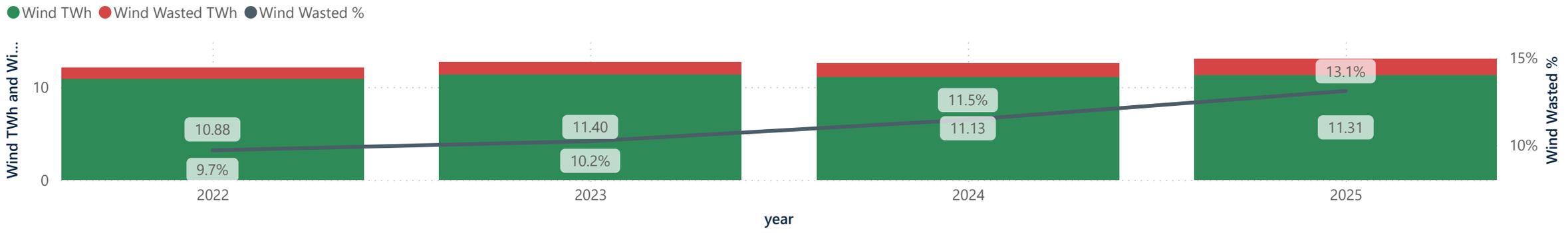
Wind Energy Wasted (TWh)

5.61

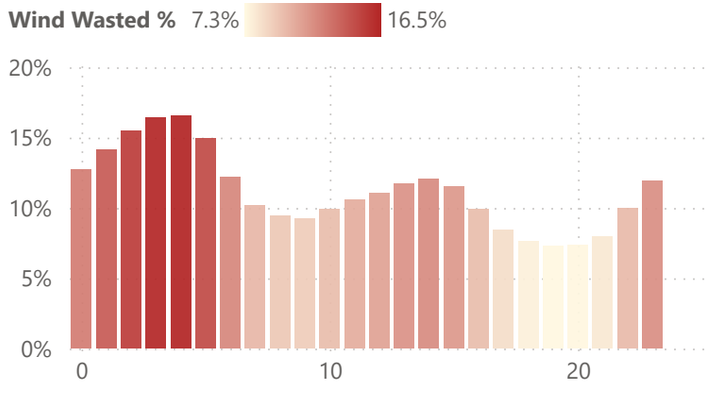
Hours Wind Supplied 50%+

9,841

Wind Energy Used vs Wasted, and Waste Rate



Wind Waste Rate by Hour of Day



full_date	Wind Share %	Demand GWh
23 February, 2025	84.6%	93.1
3 February, 2025	82.2%	96.0
26 February, 2022	81.5%	89.2
26 November, 2022	81.0%	87.9
5 February, 2022	80.9%	91.3
10 November, 2022	78.9%	92.4
20 February, 2025	78.5%	101.3
21 December, 2024	78.4%	95.9
28 December, 2023	76.8%	90.5
5 October, 2024	76.2%	87.6

2022–2025: The share of available wind energy that was wasted rose every year, from 9.7% to 13.1%. Wind is mostly wasted at midday (overlapping with solar) and overnight (low demand), and least wasted during the evening peak, pointing to grid constraints and the need for storage.

How Does Ireland Cope When the Wind Drops?

YEAR

2022 | 2023 | 2024 | 2025

Net Imports from GB (TWh)

8.89

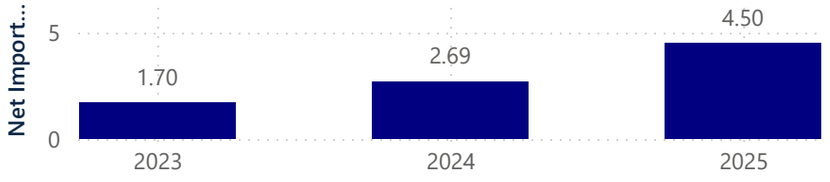
GB Demand Met by Imports

8.8%

Time Spent Importing%

84.1%

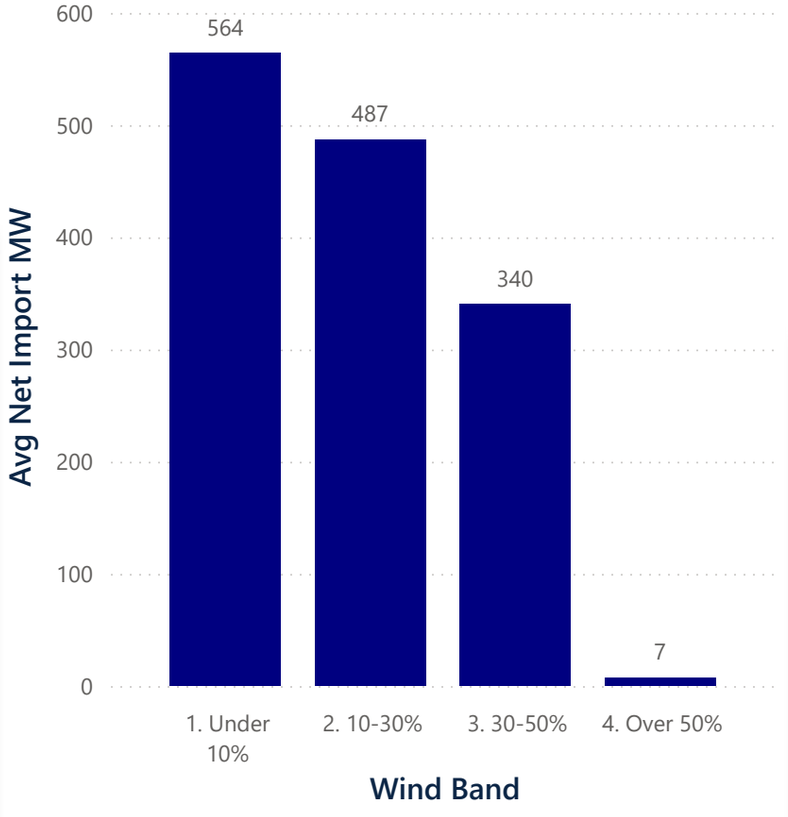
Net Electricity Imports from Great Britain (TWh)



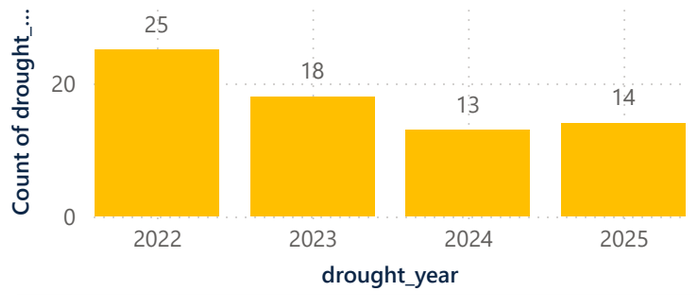
Longest Wind Droughts (wind under 10% of demand)

drought_start	drought_end	Hours	Days
8/7/2022 4:45:00 PM	8/12/2022 9:30:00 PM	125.00	5.20
8/20/2025 3:30:00 AM	8/23/2025 11:00:00 PM	91.80	3.80
9/16/2022 2:45:00 PM	9/20/2022 2:00:00 AM	83.50	3.50
5/8/2025 6:00:00 AM	5/11/2025 2:30:00 PM	80.80	3.40
3/2/2023 8:30:00 AM	3/5/2023 2:00:00 PM	77.80	3.20
11/10/2024 7:15:00 PM	11/13/2024 9:15:00 PM	74.30	3.10
4/27/2022 8:45:00 AM	4/30/2022 9:45:00 AM	73.30	3.10
1/19/2025 10:45:00 PM	1/22/2025 10:15:00 PM	71.80	3.00
6/21/2022 3:15:00 AM	6/24/2022 12:45:00 AM	69.80	2.90
4/9/2025 2:30:00 PM	4/12/2025 11:45:00 AM	69.50	2.90

Average Imports by Wind Level (MW)



Wind Droughts of 24+ Hours per Year



Ireland imports more electricity when the wind drops: about 564 MW on average when wind supplies under 10% of demand, versus almost zero above 50%. Net imports grew from 1.7 TWh (2023) to 4.5 TWh (2025). Multi-day wind droughts occur every year, with the longest lasting 125 hours (5.2 days) in August 2022.